

RONAK DUTTA

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Education

University School of Automation and Robotics (GGSIPU)

B.Tech in Industrial Internet of Things | GPA: 8.81/10

Expected Sept. 2027

New Delhi, India

Bal Mandir Sr. Sec. School

CBSE Class XII | 92.6%

2022

Delhi, India

Relevant Coursework

- Data Structures & Algorithms
- Operating Systems
- Database Systems
- Computer Networks

Experience

1M1B Green Internship – Smart Irrigation IoT System

July 2025 – Aug. 2025

Green Skills Intern

- Designed a prototype smart irrigation system using IoT devices (ESP32, soil moisture & temperature sensors) to automate water scheduling.
- Built a web-based dashboard for real-time monitoring and remote control of irrigation, enabling actionable water-use insights.
- Leveraged Tableau for sustainability dashboards, visualizing patterns and trends in environmental data.

Projects

SkillSwap – Decentralized Skill Exchange | *PostgreSQL, Express, React, Node.js, TailwindCSS* | [Live Demo](#) | [GitHub](#)

- Architected a full-stack application enabling users to monetize or barter skills via a custom virtual currency system.
- Engineered a robust REST API with Node.js and PostgreSQL to handle complex transaction logic, including wallet deductions, skill ownership transfers, and atomic swap negotiations.
- Implemented secure JWT authentication and a Handshake Protocol allowing users to initiate, accept, or reject barter requests with real-time status updates.

ATS Workplace – AI-Powered Recruitment Suite | *React, Node.js, Python, FastAPI, PostgreSQL*

- Architected a microservice-based ATS featuring a React frontend, a Node.js API gateway, and a Python-based ML engine to automate high-volume recruitment workflows.
- Developed a high-performance NLP pipeline using FastAPI, SpaCy, and Hugging Face Transformers for real-time candidate analysis and semantic resume scoring against job descriptions.
- Engineered an automated screening system that generates AI-driven candidate summaries and ranking metrics, significantly reducing manual review time for HR teams.

SafarSaathi – Cab Booking Application | *PostgreSQL, Express, React, Node.js, Twilio, TailwindCSS* | [Live Demo](#) | [GitHub](#)

- Implemented a multi-tier RBAC system managing distinct authentication flows for Admins, Drivers, and Customers via JWT.
- Developed a comprehensive Admin Dashboard with data visualization for revenue tracking and a fleet management system with real-time driver assignment and automated WhatsApp confirmations via Twilio.
- Integrated Razorpay for secure payment processing and built a complete driver onboarding workflow from application to approval, streamlining the end-to-end booking lifecycle.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, C++, SQL

Front-End: React.js, Next.js, Tailwind CSS, Redux

Back-End: Node.js, Express.js, FastAPI, Axios

Databases: MongoDB, PostgreSQL, MySQL, Firebase

Tools & Platforms: Git, GitHub, Postman, Vercel, Tableau

Certifications

React – The Complete Guide (Hooks, React Router, Redux) – Jonas Schmedtmann (Udemy)

The Complete Full-Stack Web Development Bootcamp – Dr. Angela Yu (Udemy)